

Supplementary Materials for Fiber Bragg Grating Whiskers for Bioinspired Hydrodynamic Perception on Underwater Robots

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Table S1: Nomenclature. Symbols and variables used throughout the main text and supplementary material.

Symbol	Description	Unit
<i>Whisker geometry</i>		
a	Major semi-axis of peak cross-section at whisker base	mm
b	Minor semi-axis of peak cross-section at whisker base	mm
k	Major semi-axis of trough cross-section at whisker base	mm
l	Minor semi-axis of trough cross-section at whisker base	mm
α	Inclination angle of peak cross-section	°
β	Inclination angle of trough cross-section	°
Λ	Undulation period	mm
<i>FBG signals</i>		
λ_1	Bragg wavelength of FBG 1	nm
λ_2	Bragg wavelength of FBG 2	nm
$\Delta\lambda$	FBG wavelength shift relative to baseline	nm, pm
<i>Modeling symbols</i>		
n_0	Number of discretized segments of the whisker	/
n_i ($i = 1, 2, 3, 4$)	Number of discretized segments of the bridges	/
$\mathbf{c}$	Constraining equations of the analytical model	/
Ψ	Joint variables of discretized segments	rad
$\mathbf{K}_s$	Structural stiffness matrix	$N/m, N \cdot m/rad$
$\mathbf{J}$	Jacobian matrix	/
$\mathbf{F}_i$ ($i = 1, 2, \dots, n_0$)	Externally applied loads to the whisker in wrench form	$N \cdot m, N$
$\mathbf{F}_{B,i}$ ($i = 1, 2, 3, 4$)	Interaction loads at the connection point in wrench form	$N \cdot m, N$
$\mathbf{g}_{st}$	Pose matrix	/
ξ	Joint twists of the equivalent mechanisms	/
E	Young's Modulus	GPa
μ	Poisson's ratio	/
G	Shear Modulus	GPa
I_{xx}, I_{yy}, I_p	Polar Moment of Inertia	m^4

Materials and Methods

Whisker morphology parameterization

The whisker geometry follows the biologically inspired undulated cross-sectional description introduced by Hanke et al. (4), with an additional quadratic taper imposed from base to tip. Table S1 lists the corresponding geometric parameters used in the CAD model and fabrication workflow. Peak and trough cross-sections are represented by elliptical semi-axes (a, b) and (k, l) , respectively, together with inclination angles α and β and undulation period Λ . To mimic the observed distal thinning of harbor-seal vibrissae, each semi-axis is scaled by the taper profile

$$s(n) = s_0 \left(1 - (nt_0)^2\right), \quad (\text{S1})$$

where $s_0 \in \{a, b, k, l\}$ is the corresponding base value, n indexes cross-sections from base to tip, and t_0 is the taper offset. The total whisker length used in this work was approximately 100 mm.

Table S2: Geometric design parameters of the whisker sensor. Here $s_0 \in \{a, b, k, l\}$ denotes the base value of each cross-sectional semi-axis, and n indexes cross-sections from base to tip.

Parameter	Symbol	Value
<i>Cross-section at base (following Hanke et al. (4) and Rinehart et al. (33))</i>		
Peak semi-axes	$a \times b$	1.575×0.534 mm ($a/b = 2.95$)
Trough semi-axes	$k \times l$	1.248×0.657 mm ($k/l = 1.90$)
Inclination angles	α, β	$15.27^\circ, 17.6^\circ$
Undulation period	Λ	3.5 mm
<i>Taper profile: $s(n) = s_0(1 - (n \cdot t_0)^2)$</i>		
Taper offset	t_0	0.029
Total length	L_t	~ 100 mm

Whisker fabrication and base structure

Each whisker comprises a superelastic ASTM F2063 nitinol wire core overmolded with a urethane elastomer (VytaFlex 60) to create the undulated external profile. Fabrication followed a dual-casting

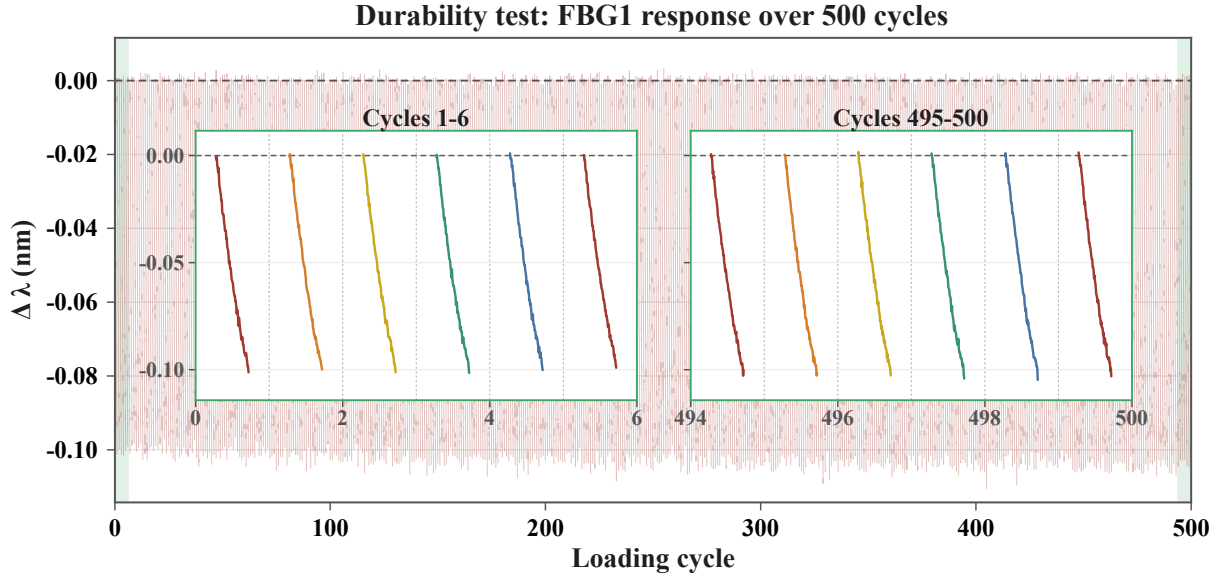


Figure S1: Durability test The preferred whisker design is subjected to 500 repeated indentation cycles on the linear stage. This plot includes a zoomed view of five representative trials showing the FBG 1 wavelength shift during loading and release and FBG 1 traces from 500 consecutive trials concatenated in time. This suggests that the whisker’s mechanical-to-optical response does not have obvious fatigue under long-term cyclic loading.

workflow. A rigid positive mold of the target whisker geometry was first printed and enclosed within a printed shell. Silicone rubber (Mold Star 16 FAST) was then cast around this positive to create a compliant negative mold. After demolding the positive, a nitinol wire was inserted into the silicone mold and urethane was injected to form the final whisker body. After curing and release, the whisker was bonded into a 3D-printed base containing two orthogonal bridge structures. A single optical fiber carrying two FBGs was routed through grooves on these bridges so that bending moments at the whisker root generate tensile or compressive strain in the gratings.

Durability and large-deflection recovery characterization

To evaluate robustness under repeated loading, the preferred whisker design was subjected to cyclic indentation on a linear stage while FBG wavelength shifts were recorded continuously (Fig. S1). A separate set of release tests examined recovery from large imposed deflections. These measurements were intended to verify that the whisker and bridge structure retained repeatable optical response after repeated loading and that the nitinol-core design recovered without obvious permanent set under the tested deflections.

To evaluate the recovery behavior of the preferred wavy-tapered whisker design under large transient deflections underwater, the whisker shaft was manually deflected and released across three trials at different contact locations along the shaft, with the maximum bending angle exceeding 90° relative to its equilibrium state. FBG 1 and FBG 2 were sampled continuously at about 2 kHz. The three trials are shown in Movie S4. Onset and recovery were extracted from each trial using the procedure below. Because the whiskers were submerged in a thermally stable water bath throughout each trial, baseline optical signals showed no measurable temperature drift on either channel and no separate temperature compensation was applied.

Force-onset detection. For each trial, we used the differential signal $\Delta\lambda_d(t) = \lambda_1(t) - \lambda_2(t)$ to detect the onset of force application, since differential transients are larger and more sharply defined than the corresponding single-channel transients during whisker bending. A provisional baseline window of length 1 s was selected at the start of the recording, and the mean $\bar{\mu}_d$ and standard deviation $\bar{\sigma}_d$ of $\Delta\lambda_d$ over this window were computed. The force-contact time t_0 was then defined as the first sample after this provisional window for which

$$|\Delta\lambda_d(t) - \bar{\mu}_d| > 10 \bar{\sigma}_d. \quad (\text{S2})$$

The factor of 10 was chosen to be robustly above the static optical-system noise floor (typically $\bar{\sigma}_d \approx 1$ pm) and well below the bending-induced peak excursions of several hundred to a few thousand picometers. With t_0 established, the time axis was shifted so that $t = 0$ corresponds to force contact, and a refined 1 s baseline window $\mathcal{B} = [-2 \text{ s}, -1 \text{ s}]$ was extracted for downstream recovery analysis. Per-channel baseline statistics $\bar{\lambda}_{\text{bl},c}$ and $\sigma_{\text{bl},c}^2$ ($c \in \{1, 2\}$) were computed over $\mathcal{B}$ and used in the recovery analyses below.

Peak excursion. The peak excursion was extracted on the bending-dominant channel (λ_1) as the largest absolute deviation from its baseline mean within an 8 s window after t_0 :

$$|\Delta\lambda|_{\text{max}} = \max_{0 \leq t \leq 8 \text{ s}} |\lambda_1(t) - \bar{\lambda}_{\text{bl},1}|, \quad (\text{S3})$$

attained at time t_{peak} .

Amplitude-based recovery: $\pm 5\%$ peak band. We first quantify how rapidly the signal magnitude returns toward baseline by reporting the time at which λ_1 re-enters and remains within $\pm 5\%$ of the peak excursion after reaching its peak. Specifically, $t_{5\%}$ is defined as the earliest time $t \geq t_{\text{peak}}$ for which

$$|\lambda_1(t') - \bar{\lambda}_{\text{bl},1}| \leq 0.05 |\Delta\lambda|_{\text{max}} \quad \text{for all } t' \in [t, t + 1 \text{ s}], \quad (\text{S4})$$

where t_{peak} is the time at which $|\Delta\lambda|_{\text{max}}$ is attained. This metric captures the practical timescale on which the sensor returns to a near-baseline reading and is reported in Fig. 2D as a shaded $\pm 5\%$ band.

Statistical recovery: sliding-window F-test. A second criterion identifies when the post-release noise level has recovered to that of the pre-contact baseline. Beginning at t_0 , a sliding analysis window of length $T_w = 5 \text{ s}$ was advanced in steps of $\Delta t = 0.02 \text{ s}$ along λ_1 . For each window starting at time t_w , the sample variance $\sigma_w^2(t_w)$ was computed and a two-sided F-test of variance equality was performed against the refined baseline:

$$F(t_w) = \frac{\sigma_w^2(t_w)}{\sigma_{\text{bl},1}^2}, \quad p(t_w) = 2 \min[P(F \leq F(t_w)), 1 - P(F \leq F(t_w))], \quad (\text{S5})$$

where $P(F \leq \cdot)$ is the cumulative distribution function of the F distribution with $(n_w - 1, n_{\text{bl}} - 1)$ degrees of freedom. The whisker was deemed recovered at the earliest window starting time t_{rec} for which $p(t_w) \geq \alpha$ with $\alpha = 0.05$ and for which the same condition continued to hold over at least 1 s of consecutive steps (a persistence criterion). Any residual mean offset is reported separately as

$$\Delta\lambda_{\text{res}} = \bar{\lambda}_w(t_{\text{rec}}) - \bar{\lambda}_{\text{bl},1}, \quad (\text{S6})$$

defined as the difference between the mean of λ_1 over the qualifying window and the baseline mean.

Per-trial results. The three release trials shown in Movie S4 contacted the whisker shaft at progressively more distal positions—near the base, near the middle, and near the tip—producing peak excursions $|\Delta\lambda|_{\text{max}}$ of 1643, 673, and 512 pm, respectively. The corresponding amplitude-based recovery times $t_{5\%}$ were 0.87, 0.77, and 0.50 s, and the F-test recovery times t_{rec} were 7.86, 4.84, and 3.80 s. Residual offsets $\Delta\lambda_{\text{res}}$ at t_{rec} were +0.4, +2.8, and +1.4 pm (or 0.02%, 0.42%, and 0.27% of the respective peak excursions), comparable to the pre-contact baseline standard

deviation ($\sigma_{bl,1} \approx 1.1$ pm). In all three trials, the signal returned within $\pm 5\%$ of its peak in under one second after the peak and reached a residual offset at the level of intrinsic optical-system noise within seconds.

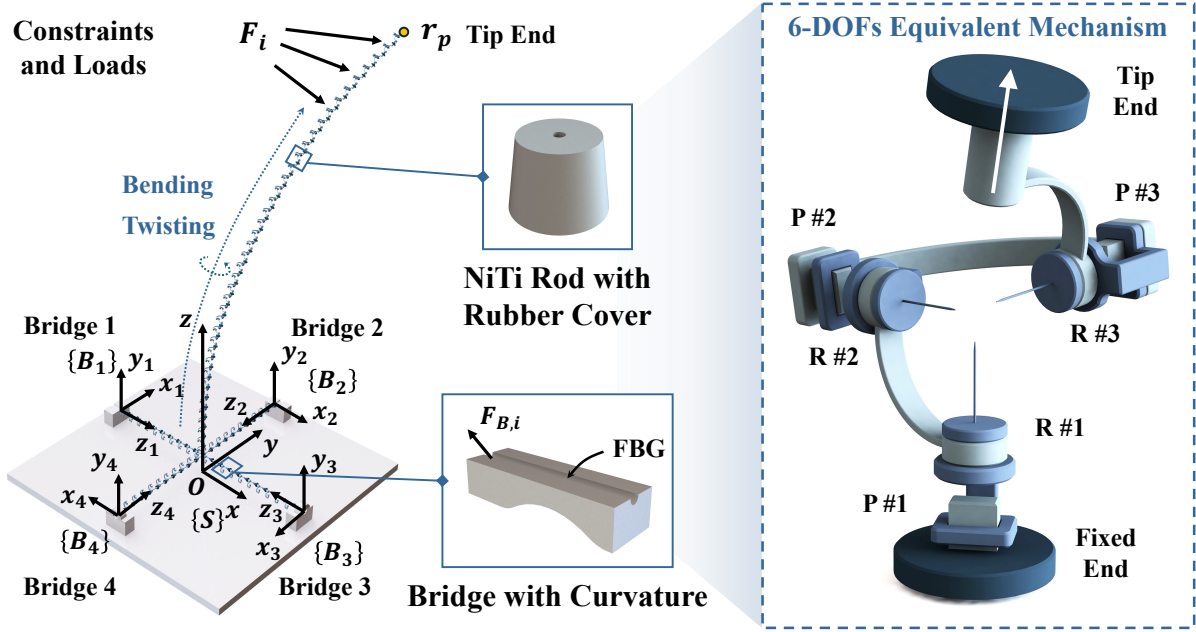


Figure S2: Analytical modeling method. Elastic units, including the whisker and the base bridge, are discretized into several segments. Each segment is equalized to a 6-DoFs serial mechanism with three translational and three rotational joints. Stiffness characteristics are assigned to each segment according to the geometry and material properties of the cross-sections. A specific frame is constructed for each component. The analysis of the elastic system is transformed into the kinematics problem of a parallel-serial hybrid mechanism.

Discretization-based analytical model

For rapid design iteration, the whisker was modeled using a discretization-based approach in which the whisker body and bridge structure were segmented into small sections and represented as an equivalent hybrid serial mechanism. Each segment was assigned stiffness terms based on its local geometry and material composition, allowing the static deformation problem to be solved as a constrained equilibrium problem under imposed displacement and contact force. In this framework, the large deflection of the whisker is captured while the bridge remains within the small-strain regime relevant to FBG transduction. The model was used to compare bridge-height and bridge-length combinations and to guide the selection of the final geometry reported in the main text.

Specifically, the whisker was discretized into n_0 relatively short segments with equal length, as demonstrated in Fig. S2. Each segment is equivalent to a 6-DoF serial mechanism with three rotational joints and three translational joints. In this case, the force-deflection behavior of the elastic unit can be represented by that of the equivalent redundant multi-body system, which is composed of rigid linkages articulated with passive joints. Further, the modeling problem is transformed into the kinematics of mechanisms, where the Product of Exponential formula can be applied to solve it in a concise and efficient manner.

As shown in Fig. S2, the spatial ground frame $\{S\}$ was constructed at point O , where the whisker is connected to the base. The x -axis and y -axis were aligned with the FBG bridge directions, and the z -axis was constructed according to the right-hand rule, which is perpendicular to the base, pointing in the direction of extension of the whisker. The FBG bridge connecting the whisker was also regarded as an elastic unit, with composite materials. The interaction between the whisker and the bridge was considered within the intersection region. In this case, the bridge was separated into four counterparts, which are circularly arranged around the whisker in a consistent manner. For each part, one end was connected to the whisker while the other end was fixed to the base. Each part of the bridge can be regarded as an elastic unit and can be modeled based on the discretization-based method by separating it into n_i relatively short segments of equal length. Furthermore, a base frame $\{B_i\}$ ($i = 1, 2, 3, 4$) was constructed at the fixed point with the y -axis in the same direction as the z -axis of $\{S\}$. To maintain consistency, the z -axis for each $\{B_i\}$ was determined as the direction pointing from the fixed point to the connected point, and the x -axis was constructed according to the right-hand rule.

Herein, the whisker sensor was kinematically equalized into a parallel-serial hybrid mechanism. The analysis of this expands around the whisker and interacts at the connected point. Suppose the externally exerted loads (containing force and torque) are given, which are distributed along the whisker. On the basis of the discretization, the load for each segment is denoted as F_i ($i = 1, 2, \dots, n_0$), which is expressed in $\{S\}$ in wrench form. The interaction forces and torques between the bridge and the whisker are represented by $F_{B,i}$ ($i = 1, 2, 3, 4$), which are unknown but correlated to the intrinsic force-deflection behavior. For the whisker, propagated torques are balanced through elastic joints. As a consequence, the kinetostatic model can be integrated into a set of nonlinear algebraic equations as Eq. (1). The stiffness components of the whisker segment can be obtained as

$$Rotational : \begin{cases} k_{x,R} \approx \frac{E_{NiTi} \cdot I_{xx,NiTi}}{\delta} \\ k_{y,R} \approx \frac{E_{NiTi} \cdot I_{yy,NiTi}}{\delta} \\ k_{z,R} \approx \frac{G_{NiTi} \cdot I_{p,NiTi}}{\delta} \end{cases} \quad Translational : \begin{cases} k_{x,T} \approx \frac{3 \cdot E_{NiTi} \cdot I_{yy,NiTi}}{\delta^3} \\ k_{y,T} \approx \frac{3 \cdot E_{NiTi} \cdot I_{xx,NiTi}}{\delta^3} \\ k_{z,T} \approx \frac{E_{NiTi} \cdot A_{NiTi}}{\delta} \end{cases} \quad (S7)$$

where $E_{NiTi} \approx 60GPa$ and $\mu_{NiTi} \approx 0.33$ are taken for NiTi material property for the rod. δ represents the length of the segment. The influence of the rubber covering is neglected due to its much smaller stiffness under the consideration of $E_{Rubber} \approx 1MPa$. It can be estimated that the stiffness provided by the rubber covering only accounts for less than 0.15%, and thus is neglected.

To solve the equations of Eq. (1), the passively adjusted forces $\mathbf{F}_{B,i}$ exerted by the bridge should be expressed according to the position and pose of the connected point. For given values of Ψ , the position and pose of the connected point are certain. This can be expressed as

$$\mathbf{g}_{st}(\Psi) = \mathbf{T}(\Psi_1) \cdot \mathbf{T}(\Psi_2) \cdot \dots \cdot \mathbf{T}(\Psi_n) \cdot \mathbf{g}_{st,0} \quad (S8)$$

where $\mathbf{T}(\cdot)$ represents the transformation matrix.

Subsequently, the separated part of the bridge can be modeled, based on both geometric and force constraints, as Eq. (2), where the stiffness component of the bridge segment can be acquired by the composition of distinct materials. It should be noted that a curve is removed at the location where the FBG sensor is arranged, leading to a minimum height of 0.45 mm. In this case, the stiffness components can be obtained accordingly.

Both valid equations and independent variables in Eq. (2) are $(6n_i + 6) \times 1$, making the model determinate. The gradient descent iteration method can be utilized to solve the equation set. The gradient of the constraining functions can be analytically derived as

$$\nabla \mathbf{c}_i = \begin{bmatrix} \mathbf{J}_t & \mathbf{0}_{6 \times 6} \\ \mathbf{K}_{s,B,i} + \mathbf{K}_{J,B,i} & -\mathbf{J}_t^T \end{bmatrix} \quad (S9)$$

The influence of the configuration is expressed in $\mathbf{K}_{J,B,i}$, which can be derived as

$$\mathbf{K}_{J,B,i} = -\frac{\partial \mathbf{J}_t^T}{\partial \mathbf{\Psi}_{B,i}} \cdot \mathbf{F}_{B,i} = - \begin{bmatrix} 0 & 0 & \dots & 0 \\ \mathbf{F}_{B,i}^T \cdot \mathbf{Z}_{\xi_1} \cdot \xi_2 & 0 & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ \mathbf{F}^T \cdot \mathbf{Z}_{\xi_1} \cdot \xi_n & \dots & \mathbf{F}^T \cdot \mathbf{Z}_{\xi_{n-1}} \cdot \xi_n & 0 \end{bmatrix} \quad (\text{S10})$$

where $\mathbf{Z}_{\xi_i} \in \mathbb{R}^{6 \times 6}$ represents the adjoint representation of joint twist ξ_i in $\{\mathbf{S}\}$.

For Newton-Raphson iteration, the increment of the variables can be obtained as

$$\begin{bmatrix} \mathbf{\Psi}_{B,i} \\ \mathbf{F}_{B,i} \end{bmatrix}_{k+1} = \begin{bmatrix} \mathbf{\Psi}_{B,i} \\ \mathbf{F}_{B,i} \end{bmatrix}_k - (\nabla \mathbf{c}_i)^{-1} \cdot \mathbf{c}_i \quad (\text{S11})$$

Eventually, the variables are gradually approaching the equilibrium states, where the whole deflection of the bridge can be reconstructed. Furthermore, the equations in Eq. (1) can also be solved based on the gradient descent method, with an analytically expressed gradient. The gradient can be expressed as

$$\nabla \mathbf{c} = \frac{\partial \mathbf{c}}{\partial \mathbf{\Psi}} = \mathbf{K}_s + \sum_{i=1}^{n_0} \mathbf{K}_{J,i} + \sum_{i=1}^4 \mathbf{K}_{F,i} \quad (\text{S12})$$

where $\mathbf{K}_{F,i} = -\mathbf{J}_c^T \cdot (\mathbf{Ad}_{\mathbf{g}_i}^T \cdot \mathbf{K}_{B,i} \cdot \mathbf{Ad}_{\mathbf{g}_i})$ represents the influence of the forces and torques interacting at the connected point. The adjoint and co-adjoint matrices express the transformation from $\{\mathbf{B}_i\}$ to $\{\mathbf{S}\}$. As a consequence, the Newton-Raphson iteration can be expressed as

$$\begin{bmatrix} \mathbf{\Psi} \end{bmatrix}_{k+1} = \begin{bmatrix} \mathbf{\Psi} \end{bmatrix}_k - (\nabla \mathbf{c})^{-1} \cdot \mathbf{c} \quad (\text{S13})$$

Eventually, both the deflections of the base bridge and the whisker can be obtained, along with the constraining forces. On this basis, parametric design of the sensing unit can be carried out with a rapid and precise model.

Processing of the geometry comparison data

For the geometry comparison, we analyzed the FBG 2 Bragg-wavelength trace from three recordings corresponding to the wavy, non-tapered, and cylindrical geometries. All recordings were acquired under the same flow condition. For each recording, time was re-referenced so that the first sample

corresponded to $t = 0$. Because the raw timestamps were nonuniform, each FBG 2 trace was linearly interpolated onto a common 1 kHz time base before filtering. A baseline offset was then removed by subtracting the mean signal over a pre-motion baseline interval, and the baseline-corrected trace was smoothed using a third-order Savitzky–Golay filter with a 31-sample window.

To ensure a fair comparison across geometries, signals were selected using the same predefined criteria rather than by independently searching for the largest peak in each recording. Specifically, each trace was first restricted to the stable-velocity portion of the gantry motion, thereby excluding the acceleration and deceleration phases. Within this stable-motion region, the same fixed-duration window was extracted from all recordings. As a result, the compared signals were matched by sensing channel, flow condition, and experimental phase. This approach avoids geometry-specific peak-picking and ensures that differences in waveform amplitude reflect a common operating regime across all geometries.

For visualization, each extracted segment was centered by subtracting its midrange, defined as $(\max + \min)/2$, so that the waveform was symmetric about zero. This centering changes only the vertical offset and does not affect the peak-to-peak amplitude.

Processing of the velocity-response data

The raw Bragg wavelength trace from channel FBG 1 was used for all runs in the wavy-whisker velocity-sweep experiment. Each trial began with an approximately 5 s stationary period before the translation started. For the figure-generation pipeline, the baseline wavelength for run r was computed from the initial static samples between $t = 0$ and 4 s, which lie entirely within that stationary interval:

$$\bar{\lambda}_{\text{base},r} = \frac{1}{N_{\mathcal{B},r}} \sum_{t_i \in \mathcal{B}_r} \lambda_r(t_i), \quad (\text{S14})$$

where $\lambda_r(t_i)$ is the measured FBG 1 wavelength in nm and $\mathcal{B}_r$ denotes the baseline window for run r .

A second window, $\mathcal{S}_r$, was then selected from the middle of the trial after the motion had reached its approximately constant-speed regime and before the final deceleration phase. The exact start and end times of this stable window were defined separately for each run to account for the different durations of the constant-speed portion at different commanded speeds. The mean wavelength in

this stable region was

$$\bar{\lambda}_{\text{stab},r} = \frac{1}{N_{S,r}} \sum_{t_i \in S_r} \lambda_r(t_i), \quad (\text{S15})$$

and the baseline-referenced response for that run was defined as

$$\Delta\lambda_r = \bar{\lambda}_{\text{stab},r} - \bar{\lambda}_{\text{base},r}. \quad (\text{S16})$$

No temporal filtering or resampling was applied for this panel; the averages were computed directly from the acquired samples within each window.

Five replicate runs were analyzed at each commanded speed (10, 20, 30, 40, 50, and 60 cm s⁻¹). Within each speed bin, the run-level values $\Delta\lambda_r$ were screened with a robust median absolute deviation criterion (modified z -score threshold $|z| \leq 4$) to remove outliers before averaging. The plotted mean response at speed v was then

$$\mu_{\Delta\lambda}(v) = \frac{1}{n_v} \sum_{r \in \mathcal{R}_v} \Delta\lambda_r, \quad (\text{S17})$$

with error bars given by the standard deviation across the retained replicates,

$$\sigma_{\Delta\lambda}(v) = \sqrt{\frac{1}{n_v - 1} \sum_{r \in \mathcal{R}_v} (\Delta\lambda_r - \mu_{\Delta\lambda}(v))^2}. \quad (\text{S18})$$

For display, the grouped mean and standard deviation were converted from nm to pm. A second-order polynomial was fit to the grouped mean response for visualization,

$$\Delta\lambda_{\text{fit}}(v) = av^2 + bv + c, \quad (\text{S19})$$

where v is the commanded speed.

Processing of the data for angle-of-attack experiment

For the angle-of-attack comparison of the wavy whisker at 60 cm s⁻¹, we analyzed seven predefined recordings acquired at 0°, 15°, 30°, 45°, 60°, 75°, and 90°. For each recording, both sensing channels (FBG 1 and FBG 2) were processed identically. The time axis was first re-referenced so that the first sample corresponded to $t = 0$. Because the raw acquisition timestamps were nonuniform, each trace was linearly interpolated onto a common uniform grid at 1 kHz. Each resampled trace was then smoothed with a third-order Savitzky–Golay filter using a 31-sample window, corresponding to a 31 ms smoothing window on the uniform grid.

Signals were selected for comparison using fixed timing criteria rather than by searching each recording for a local maximum response. Each recording was first restricted to its predefined gantry stable-motion interval, which excluded the acceleration and deceleration phases of the traverse. Within this stable-motion interval, the same absolute-time window was extracted from every angle-of-attack condition and from both FBG channels. Thus, the compared signals were taken from the steady-velocity portion of the traverse, but they were not chosen by a midpoint-seeking algorithm; instead, they were taken from a common predefined operating window that lay well within the stable-motion region for all runs. Using the same absolute-time window across all recordings enabled direct comparison of waveform shape and amplitude across angle-of-attack conditions.

For visualization, each extracted segment was centered independently by subtracting its midrange, defined as one-half of the sum of the segment maximum and minimum values, so that the oscillatory waveform was displayed about zero. This centering changed only the vertical offset and did not alter the waveform shape or peak-to-peak amplitude.

Processing of the data for static pitching foil animation

Each run of the scanning matrix contains a synchronized time series with two FBG channels and a motor-angle command, together with synchronized gantry telemetry giving the sensor position in the foil frame. For each run, the first valid synchronized sensor-relative position was used as the fixed spatial location associated with that run.

For signal processing, the sampling frequency of each run was estimated from the recorded timestamps, and the two FBG channels were linearly interpolated onto a uniform time grid at that run-specific sampling rate. In the packaged dataset used here, no separate per-run timing-annotation files were present; therefore, the DC baseline of each FBG channel was defined as the mean of the first 1000 resampled samples and subtracted from the full trace. The baseline-corrected signals were then filtered with a zero-phase fourth-order Butterworth low-pass filter with a 10 Hz cutoff. No additional Savitzky–Golay smoothing or band-pass filtering was applied. The processed traces were then interpolated again onto a common animation time base sampled at 20 frames s^{-1} , and only the interval from $t = 4 \text{ s}$ to $t = 28 \text{ s}$ was rendered.

At each spatial location, the processed FBG_1 and FBG_2 signals were used as the horizontal and

vertical components of a 2-D response vector. The displayed signal magnitude was computed as

$$M(t) = \sqrt{\text{FBG}_1(t)^2 + \text{FBG}_2(t)^2}.$$

Vector length was scaled so that the 99.5th percentile of M across the full dataset corresponded to 60 mm in the plot. Vector color encoded the unscaled magnitude using the `turbo` colormap with a fixed normalization range of 0 to 0.015 nm. To suppress pre-motion baseline noise, vectors were masked before the inferred onset of motor motion and outside the valid recording interval for each run.

The foil was rendered as a rigid line of length 151.56 mm rotating about a fixed pivot. Because the packaged CSV files contain a synchronized commanded motor-angle trace but not a measured motor-angle trace, the animation used `target_position.rad` to update foil orientation at each frame. The final movie was rendered in Matplotlib at 20 frames s^{-1} and exported as an MP4 file using `ffmpeg`. Still frames were exported from the same rendering pipeline by selecting the nearest animation frame to a requested timestamp and saving raster and vector versions.

Processing of data for the moving-source trail analogue animation

The moving-source animation was generated from synchronized FBG and gantry telemetry recorded for each run. For each run, the time axis was re-zeroed to the first valid sample, and the two FBG channels were baseline-corrected by subtracting the mean of the first 1000 samples. The commanded foil angle was taken from the motor target trajectory. In the final configuration used for the supplementary animation, no low-pass filtering was applied so that transient peak responses were preserved. To reduce run-to-run timing jitter, all runs were aligned by the onset of foil motion, defined as the first time at which the commanded angle deviated from its initial value by at least $\max(0.02 \text{ rad}, 0.1A_{\text{peak}})$, where A_{peak} is the peak angular excursion in that run. Each run was then shifted so that this onset coincided with the median onset time across all runs.

Because the travel range of the small gantry did not allow both sides of the measurement domain to be sampled in a single acquisition sequence, the full matrix was assembled from two batches acquired with opposite initial foil phases. In the first batch, the foil started near one peak angle ($\approx +0.4 \text{ rad}$); in the second batch, it started near the opposite peak angle ($\approx -0.4 \text{ rad}$). The second batch was mapped onto the opposite side of the domain by reflecting the sensor

coordinates about the trajectory centerline according to $y' = 2y_c - y$, with $y_c = 625$ mm. After this geometric reflection, the transverse component used in the animation (FBG_2) was sign-inverted so that vector directions remained consistent under reflection. This reconstruction, therefore, assumes approximate symmetry of the response field about the trajectory centerline when the foil is started from the opposite peak angle.

After alignment and mirroring, the sampled positions were snapped to a regular grid with 16 mm spacing in x and 4 mm spacing in y . The FBG signals and the foil-pivot trajectory were resampled onto a common 20 Hz time grid using linear interpolation, with edge values held outside the sampled interval, and the animation was rendered over $t = 3$ to 11 s. At each grid location, the vector components were taken as $(\text{FBG}_1, \text{FBG}_2)$, with FBG_1 plotted along the towing direction and FBG_2 plotted along the transverse direction. The color at each location represented the signal magnitude,

$$M(t) = \sqrt{\text{FBG}_1(t)^2 + \text{FBG}_2(t)^2}.$$

The sensor grid was displayed in a foil-following frame so that the matrix translated with the moving foil. Arrow lengths were scaled for visualization on a frame-by-frame basis, whereas color provided the consistent magnitude encoding across the animation.

Wake-generator kinematics and Strouhal-number convention

The wake experiments compare two source conditions: a fixed pitching foil in quiescent water and a co-translating pitching foil operating at finite St . For the co-translating case, the foil and whisker were translated together at $U = 0.46$ m/s while the foil pitched with angular amplitude $A_\theta = 0.4$ rad and frequency $f = 1.2$ Hz. The Strouhal number was defined from the foil kinematics as

$$St = \frac{f A_{\text{TE}}}{U}, \quad (\text{S20})$$

where A_{TE} is the trailing-edge excursion and U is the translation speed. Under the operating point used in the robot experiments, this yields $St = 0.35$, selected to move away from the quiescent-water limit and toward a regime more relevant to the hydrodynamic trail of a moving source.

Whisker-signal classifier for branch selection

Task geometry and delay. For the delayed branch-selection task in Fig. 6D, the robot’s start point was approximately 0.40 m behind the foil’s initial position along the nominal path. We estimated that about 5 s elapsed between foil-motion onset and the robot reaching that position. This value is reported as an operational benchmark for delay in the task geometry, not as an exact age of the fluid structures later sampled in the downstream decision region.

Task definition and labels. For the branch-selection experiment in Fig. 6, we trained a binary classifier to distinguish straight continuation from turn selection using only the two whisker signals (FBG 1 and FBG 2). Straight continuation was encoded as label 0 and turn selection as label 1. Ground-truth labels were read directly from the offline and online manifest files and were not recomputed from thresholds at runtime. No visual-tracking signals, robot pose, velocity, or yaw measurements were provided to the model.

Window extraction and split protocol. Each raw whisker record contained three synchronized columns, `time_seconds`, `fbg_1`, and `fbg_2`, with the local time origin reset to zero for each record. Offline sample timestamps were provided in the sample manifest at 40 Hz. For each manifest row, we extracted a trailing 2.0 s window ending at the listed timestamp. Because adjacent windows overlapped strongly, the train/validation/test partition was performed at the record (trajectory) level before window extraction to avoid leakage across splits. With this fixed split, the offline dataset contained 763 windows in total: 463 training windows, 114 validation windows, and 186 held-out test windows. Online evaluation used one first-decision window per run, so the first prediction became available after the first 2.0 s of whisker data.

Preprocessing. Within each 2.0 s window, both FBG channels were linearly resampled to a uniform 500 Hz grid (1000 samples per channel). When a requested window extended slightly before the beginning of a record, the missing prefix was padded with the earliest available sample value from that record. Each window was then detrended, mean-centered, and low-pass filtered with a sixth-order Butterworth filter at 25 Hz using zero-phase forward–backward filtering. No additional high-pass, band-pass, or Savitzky–Golay smoothing was applied.

Feature extraction. From each preprocessed window we computed a 42-dimensional engineered feature vector, consisting of 21 features from FBG 1 and the same 21 features from FBG 2. For each channel, four autocorrelation features summarized the peak normalized autocorrelation over positive lags within the decision window, the lag at that peak, the ratio of that peak to the mean absolute autocorrelation over the same lag range, and the positive-lag autocorrelation energy. Six frequency-domain features were obtained from Welch power spectral density estimates: band powers in five prespecified sub-bands spanning the retained 0.5–25 Hz range, together with the dominant frequency below 25 Hz. To capture temporal growth within the decision window, the 2.0 s window was divided into four equal segments and 11 temporal-segment features were computed: the RMS and mean of each segment, plus the ratios of successive segment RMS values ($\text{seg}_2/\text{seg}_1$, $\text{seg}_3/\text{seg}_2$, and $\text{seg}_4/\text{seg}_3$). No explicit cross-channel correlation features were included in the submitted classifier.

Classifier. The 42-dimensional feature vector was standardized with a `StandardScaler` fit on the training set only. We then trained a logistic-regression classifier with an ℓ_1 penalty,

$$p(y = 1 \mid \mathbf{x}) = \sigma(\mathbf{w}^\top \tilde{\mathbf{x}} + b), \quad \sigma(z) = \frac{1}{1 + e^{-z}}, \quad (\text{S21})$$

where $\tilde{\mathbf{x}}$ denotes the standardized feature vector. The fixed hyperparameters used in the submitted model were `solver=liblinear`, `penalty=l1`, `C=0.1`, `max_iter=1000`, `fit_intercept=True`, and `random_state=2`. A window was assigned to the turn class when the predicted probability satisfied $p(y = 1 \mid \mathbf{x}) \geq 0.5$; otherwise, it was assigned to the straight class. No class reweighting or post hoc threshold tuning was applied.

Performance and feature sparsity. On held-out offline test windows, the classifier achieved 77.4% accuracy, an AUROC of 0.863, and an average precision of 0.776. In the online first-decision evaluation, the same frozen scaler and classifier were applied without retraining, using a 2.0 s window per run. Across 20 online runs, the classifier selected the correct branch in 17 cases (85.0%); representative runs are shown in Movie S9. Because an ℓ_1 penalty was used, the final model was sparse: 16 of the 42 candidate features had nonzero coefficients. The retained features included both low-to-mid-frequency spectral terms and temporal-segment descriptors from both FBG channels, consistent with the progressive signal buildup seen in turning trials in Fig. 6C. Because adjacent

offline windows overlapped strongly, these window-level metrics should be interpreted as classifier-development diagnostics; the run-level online evaluation is the more behaviorally relevant measure of task performance.

Notes on wake-generator regimes

The wake experiments in the main text compare a fixed pitching foil in quiescent water with a co-translating pitching foil operating at finite St . This distinction is important because the biological inspiration for the work is delayed trail following behind a moving prey item, rather than generic response to any unsteady upstream disturbance. Prior fluid-mechanics studies reported that a rigid symmetric foil pitching in quiescent water produces a laterally meandering jet rather than a stable downstream-convected reverse von Kármán wake (40, 41). We therefore use the co-translating condition at $St = 0.35$ as the task-relevant operating point for the robotic branch-selection experiments, consistent with the commonly cited efficiency-associated range for oscillatory propulsion and thrust-producing wakes (42, 43).

Notes on literature comparison

For the flow-speed column, we report nonzero steady-flow speed(s) used for calibration or validation, rather than copying a lower bound of 0 m/s. A 0 m/s recording is a static or no-flow baseline and is useful for offset, noise, and signal-to-noise calculations, but it is not by itself evidence of a minimum measurable flow speed. When the individual positive steady-flow speeds are available, we list the smallest and largest positive speeds. When a source paper reports a range such as $0-v_{\max}$ but does not make the smallest positive tested speed recoverable, we report only ‘ $\max v_{\max}^{\dagger}$ ’ in the table. This convention avoids implying that a sensor has been calibrated down to arbitrarily small nonzero velocities. We also distinguish steady-flow calibration from oscillatory-flow or dipole-stimulus velocity measurements; the latter are retained in the modality-specific metrics table when relevant, but are not used to populate the nonzero steady-flow column.

Tables S3–S6 provide the extended cross-study comparison supporting Table 1. Table S3 expands the literature set beyond the representative studies retained in the main text and includes additional underwater whisker sensors that are historically relevant or less directly comparable to

Table S3: Extended comparison of underwater whisker-inspired sensors. This table expands the main-text comparison to include additional studies and the rows not retained in the representative comparison. The flow column reports nonzero steady-flow speed(s) used for calibration or validation. A zero-flow recording is treated as a static/no-flow baseline, not as the lower bound of a measurable flow-speed range. Entries marked [†] indicate cases where the source paper reports a range beginning at zero or includes a no-flow baseline, but the smallest positive steady-flow speed was not unambiguously recoverable; for those rows, only the maximum reported/tested steady-flow speed is listed. When a wake distance is also shown in the normalized form d_w , that notation follows the study-specific whisker-diameter convention used in the source paper or reproduced here for context.

Reference	Principle	$L \times d_1 \times d_2$ (mm)	Nonzero steady-flow speed(s) (m/s)	Wake source	Wake distance	System-level demonstration
Stocking/Eberhardt (7, 8)	Capacitance	$40 \times 2 \times 2$	max 0.88 [†]	N/A	N/A	Steady/oscillatory flow sensing
Alvarado et al. (9–11)	Resistive	$170 \times 3 \times \text{N/A}$	0.1–0.3	Cylinder	N/A	Steady flow and direction sensing
Gul et al. (20)	Resistive	$160 \times 8 \times 8$	N/A	Fishtail	N/A	Wake detection
Beem et al. (15, 16)	Resistive	$275 \times 35.7 \times 14.4$	0.2–0.8	Cylinder	$160 d_w$	Benchtop wake sensing
Liu et al. (17)	Resistive	$35 \times 2.14 \times 1.06$	0.048–0.25	Cylinder	250 mm ($\sim 235 d_w$)	Benchtop wake sensing
Zheng et al. (18, 19)	Resistive	$150 \times 11.6 \times 5.0$	0.1	Cylinder	80 mm ($\sim 10 d_w$)	Benchtop wake sensing
Kottapalli et al. (12, 13)	Piezoelectric	$3.96 \times 1.14 \times 0.52$	N/A	N/A	N/A	MEMS flow sensing
Zhang et al. (14)	Piezoelectric	$6.55 \times 1.9 \times 0.92$	N/A	N/A	N/A	Angle-of-attack and oscillatory-velocity sensing
Guo et al. (44)	Piezoelectric	$210 \times 15 \times 7.5$	0.25–0.5	Cylinder	240 mm ($\sim 16 d_w$)	Benchtop wake sensing
Wang et al. (21, 22)	TENG	$150 \times \text{N/A} \times \text{N/A}$	N/A	Fishtail	750 mm	Benchtop wake sensing
Xu et al. (27, 28)	TENG	$200 \times 12.4 \times 5.0$	max 0.6 [†]	Cylinder	40 mm	ROV state estimation
Dai et al. (23)	Magnetic	$10.92 \times \text{N/A} \times \text{N/A}$	max 0.6 [†]	Fishtail	150 mm ($\sim 95 d_w$)	Array sensing
Elshalakani et al. (24)	Optical	$100 \times 0.75 \times 0.75$	0.3	Cylinder	875 mm	Wake-source localization
Glick et al. (25)	FBG	$80 \times 0.25 \times 0.25$	0.1–0.25	Cylinder	750 mm	Benchtop wake sensing
Wang et al. (26)	FBG	$130 \times 7.14 \times 2.88$	max 0.36 [†]	Fishtail	300 mm ($\sim 95 d_w$)	Benchtop wake sensing
Our design	FBG	$100 \times 3.15 \times 1.068$	0.1–0.6	Moving foil	576.44 mm ($\sim 484 d_w$)	online branch selection

the wake-tracking robot task emphasized here. Table S4 collects modality-specific quantities that were removed from the main-paper table, including directional response, imposed oscillatory-flow response, sensitivity, response time, and signal-to-noise ratio (SNR). These quantities are useful to archive, but they are not emphasized in the main text because they are reported unevenly across the literature and are often defined differently across studies.

Where a metric was not explicitly tabulated by the source paper but could be extracted transparently from a plotted calibration, a simple local slope, or a closely matching secondary summary, we report it only as an approximate estimate, marked by a leading $\sim$ in Table S3. These values are included only for context and are not treated as directly reported author values in the main-text comparison.

Sensitivity values are reproduced in the original units and, more importantly, with the original definitions used by each study. This is critical across sensing modalities, because the reported

quantity may be capacitance change, normalized resistance change, voltage, pressure sensitivity, or wavelength shift rather than the same physical transfer function. Even within FBG studies the protocols are not identical. Wang et al. define sensitivity from the derivative of a calibrated polynomial relating average wavelength shift to steady-flow velocity and report a maximum of $0.86 \text{ nm}/(\text{m}\cdot\text{s}^{-1})$, whereas Glick et al. define sensitivity as the weakest perceivable disturbance and emphasize disturbance-tracking accuracy and SNR rather than a dedicated steady-flow wavelength-shift calibration (25, 26). For Glick et al., Fig. 9 suggests an approximate local wavelength–velocity slope of $\sim 0.12 \text{ nm}/(\text{m}\cdot\text{s}^{-1})$ over $0.25\text{--}0.30 \text{ m/s}$, but we cite that number only as a figure-derived estimate because it is not the authors’ primary sensitivity definition (25). Our reported $0.23 \text{ nm}/(\text{m}\cdot\text{s}^{-1})$ should therefore be interpreted as the steady-flow wavelength-shift calibration of our packaged whisker under towing experiments, not as a universal ranking across all optical or all whisker sensors. Within the narrower FBG subset, our value is numerically intermediate between Glick et al.’s approximate figure-derived local slope of $\sim 0.12 \text{ nm}/(\text{m}\cdot\text{s}^{-1})$ and Wang et al.’s reported maximum derivative-based value of $0.86 \text{ nm}/(\text{m}\cdot\text{s}^{-1})$, but we treat that ordering cautiously because the calibration definitions and sensor architectures are not identical (25, 26). Relative to the orthometric-FBG whisker of Wang et al. ($130 \times 7.14 \times 2.88 \text{ mm}$), our whisker ($100 \times 3.15 \times 1.068 \text{ mm}$) is 23% shorter, 56% smaller in the larger cross-flow dimension, and 63% smaller in the smaller cross-flow dimension. By contrast, Glick et al. use bare silica fibers of $80 \times 0.25 \times 0.25 \text{ mm}$, placing that design in a materially different structural regime.

For dimensions, we report each whisker as $L \times d_1 \times d_2$ when the corresponding quantities can be recovered from the source paper. For cylindrical sections, $d_1 = d_2$; for elliptical sections, d_1 and d_2 denote the major and minor cross-flow dimensions; and for undulated whiskers, we use the reported peak cross-sectional dimensions when available, consistent with the geometric description commonly used for seal-whisker-inspired morphologies (4). When a paper reports only an average diameter or uses different dimensions across different experiments, we retain the values most relevant to the reported flow or wake experiment and note that choice in Table S4.

Wake-detection distance is especially criterion-dependent. Across the literature, the reported value can denote (i) the farthest source–sensor spacing at which the dominant shedding frequency is recovered within a prescribed error band, (ii) the farthest tested spacing used for disturbance-tracking or localization experiments, or (iii) simply the largest validated point within a bounded set

of tested source–sensor separations for a particular wake generator and flow condition (17, 25–27). When source papers additionally normalize wake distance by d_w , we retain that notation because several studies use d_w to denote a study-specific whisker mean cross-flow diameter or equivalent effective-diameter convention, not the wake-generator diameter. For example, Liu et al. used a 5 cm upstream cylinder and reported 250 mm as the largest tested distance for which the dominant wake frequency remained within a 10% error margin, whereas Glick et al. evaluated upstream-cylinder spacings from $10D_c$ to $25D_c$ in disturbance-tracking experiments rather than defining a single universal wake-limit criterion. Wang et al. reported 300 mm as the farthest tested longitudinal position before the wake signal was inferred to fall below a useful SNR beyond the measured range, and Xu et al. reported 40 mm from the largest tested cylinder–sensor spacing in their wake experiment (17, 25–27). These values also depend strongly on wake-generator class, source size and kinematics, and background flow speed, so they should be interpreted as study-specific validation distances rather than universally comparable limits of whisker performance.

For our design, the reported 576.44 mm was obtained in the co-translating experiments, where the whisker and pitching foil were towed simultaneously. The quoted value corresponds to the farthest source–sensor separation sampled on our positioning gantry under that specific wake regime. We therefore report it as the maximum validated distance within our available experimental workspace, not as an architecture-independent upper bound on wake-detection range.

Response time is reported only when the source paper gives an explicit timing metric, but these values are not standardized across studies. For example, Gul et al. report a response time of about 220 ms from manual whisker bending of approximately 5 mm rather than from a synchronized wake-contact measurement (20). Xu et al. report a response time of 21 ms, but the main text attributes this value to Supplementary Fig. S7 without specifying the exact timing procedure in the main paper (27). In our study, response time is defined as

$$\Delta t_{\text{resp}} = t_{\text{FBG, onset}} - t_{\text{contact}},$$

where t_{contact} is the first frame in synchronized 240 Hz video for which physical contact with the whisker is visually observable, and $t_{\text{FBG, onset}}$ is the first synchronized interrogator sample that clearly departs from the pre-contact baseline. The reported 29.74 ms therefore represents end-to-end latency from visually resolved contact to measurable FBG output, with temporal resolution

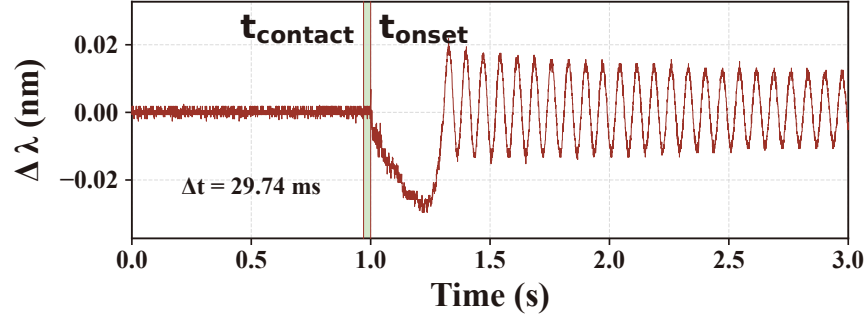


Figure S3: Response-time characterization. The end-to-end sensing latency of the preferred whisker design was measured by synchronizing video-resolved contact with the FBG 1 wavelength-shift signal. The contact time, t_{contact} , was defined as the first video frame in which physical contact with the whisker was visually observed, and the signal onset, $t_{\text{FBG,onset}}$, was defined as the first synchronized FBG sample that clearly departed from the pre-contact baseline. The response time was calculated as $\Delta t_{\text{resp}} = t_{\text{FBG,onset}} - t_{\text{contact}}$. For the representative trial shown, the measured response time was 29.74 ms, indicating rapid conversion of whisker contact into a measurable optical response.

partly limited by the video frame period (4.17 ms), as shown in Fig. S3.

Signal-to-noise ratio (SNR) is also definition-dependent and, in several studies, the exact calculation is not fully specified. Glick et al. compute SNR from the disturbance-induced wavelength shift relative to the maximum background fluctuation in wavelength, whereas Wang et al. report about 19 dB under a specified wave-noise condition ($D/L = 1$, $f = 2$ Hz, and $\theta = 54^\circ$) without giving the same numerator/denominator definition in the main text (22, 25). Wang et al. further report wake SNR values decreasing from 7.22 to 1.24 dB with longitudinal distance, again without explicitly defining the exact SNR formula in the main text, and Xu et al. report 61.66 dB but defer the procedure to Supplementary Fig. S7 (26, 27). In our work, SNR is computed from the steady-flow sensing experiment using a mean-shift detectability metric,

$$S = |\mu_{\text{steady}} - \mu_{\text{static}}|, \quad N = \sigma_{\text{static}},$$

and, in decibels,

$$\text{SNR}_{\text{dB}} = 20 \log_{10} \left(\frac{S}{N} \right).$$

This definition measures how clearly the steady-flow-induced mean shift separates from the no-flow baseline fluctuations. Our SNR should therefore not be treated as numerically interchangeable with event-based, peak-based, or maximum-fluctuation SNR values reported elsewhere.

Table S4: Modality-specific comparison metrics for underwater whisker-inspired sensors.

Shape codes: C = cylindrical, U = undulating/wavy, T = tapered; combined codes indicate multiple shape features, e.g., UT = undulating and tapered. Sensitivity, response time, and SNR are listed in the units and definitions used by the source papers when directly reported. A leading ~ denotes an approximate value estimated from a plot, a local slope, or a closely matching secondary summary when the source paper did not explicitly tabulate the metric. Because disturbance type, calibration procedure, and signal definitions differ across studies, these quantities are intended for cautious within-modality interpretation rather than as universally comparable figures of merit.

Reference	Shape	Directional discrimination	Oscillatory flow	Sensitivity (study units)	Response time	SNR
Stocking/Eberhardt (7, 8)	C	Yes	Yes	$\sim 3.55 \text{ pF}/(\text{m}\cdot\text{s}^{-1})$	N/A	N/A
Alvarado et al. (9–11)	UT	Yes	No	$\sim 1.3 \text{ V}/(\text{m}\cdot\text{s}^{-1})$	N/A	N/A
Gul et al. (20)	C	Yes	No	N/A	220 ms	N/A
Beem et al. (15, 16)	U	Yes	No	N/A	N/A	N/A
Liu et al. (17)	U	Yes	Yes	$\sim 42.5\%/(\text{m}\cdot\text{s}^{-1})$	N/A	N/A
Zheng et al. (18, 19)	UT	Yes	No	N/A	N/A	N/A
Kottapalli et al. (12, 13)	U	Yes	Yes	$2.314 \text{ V}/(\text{m}\cdot\text{s}^{-1})$	N/A	N/A
Zhang et al. (14)	C	Yes	Yes	$1.445 \text{ V}/(\text{m}\cdot\text{s}^{-1})$	N/A	N/A
Guo et al. (44)	U	Yes	No	0.01796 V/kPa	N/A	N/A
Wang et al. (21, 22)	U	Yes	No	N/A	N/A	19 dB
Xu et al. (27, 28)	C	Yes	Yes	$1.16 \text{ V}/(\text{m}\cdot\text{s}^{-1})$	21 ms	61.66 dB
Dai et al. (23)	U	Yes	No	N/A	N/A	N/A
Elshalakani et al. (24)	C	Yes	No	N/A	N/A	N/A
Glick et al. (25)	C	Yes	No	$\sim 0.12 \text{ nm}/(\text{m}\cdot\text{s}^{-1})$	N/A	2 dB
Wang et al. (26)	U	Yes	Yes	$0.86 \text{ nm}/(\text{m}\cdot\text{s}^{-1})$	N/A	1.24–7.22 dB
Our design	UT	Yes	Yes	$0.23 \text{ nm}/(\text{m}\cdot\text{s}^{-1})$	29.74 ms	39.44 dB

Table S5: Conventions used for the cross-study comparison: general quantities. This table records the normalization and reporting choices used in the representative comparison together with Tables S3 and S4.

Quantity	Convention used in this paper
Dimensions	Reported as $L \times d_1 \times d_2$ when recoverable from the source paper. For cylindrical sections, $d_1 = d_2$. For elliptical sections, d_1 and d_2 denote the major and minor cross-flow dimensions. For undulated whiskers, peak cross-sectional dimensions are used when authors report semi-axes or equivalent geometry.
Nonzero steady-flow speed(s)	Reported as the smallest and largest positive steady-flow speed used for calibration or validation when those values are recoverable. A 0 m/s recording is treated as a static/no-flow baseline, not as the lower bound of a measurable flow-speed range. Entries marked [†] indicate cases where the source paper reports a range beginning at zero or includes a no-flow baseline, but the smallest positive steady-flow speed was not unambiguously recoverable; for those rows, only the maximum reported/tested steady-flow speed is listed. Oscillatory-flow and dipole-stimulus velocity measurements are not used to populate this column.
Wake distance	Reported in absolute units when given by the source paper. When available, the distance is also expressed relative to the study-specific whisker mean cross-flow diameter d_w or equivalent effective-diameter convention used by the source paper. Because different studies use different wake generators, source kinematics, background flow speeds, and detection criteria, these values are interpreted as study-specific validation distances rather than universally comparable figures of merit. Our reported value is the farthest co-translating source–sensor separation sampled on the available positioning gantry.
Missing values	Entries are listed as N/A when the corresponding quantity was not explicitly reported, when it was not recoverable without ambiguous figure extraction, or when the reported quantity corresponds to a different stimulus class than the column definition.

Table S6: Conventions used for the cross-study comparison: performance-metric reporting. This table complements Table S5 by summarizing how sensitivity, response time, SNR, and approximate values are handled across studies.

Quantity	Convention used in this paper
Directional discrimination	Marked “Yes” when the design or results show directional sensitivity, including angle-of-attack tests, channel sign changes, or left/right wake localization.
Oscillatory flow	Marked “Yes” only when imposed oscillatory or periodic flow is explicitly tested, not merely when a vortex-generator wake is used.
Sensitivity	Reproduced in each study’s original units and definitions. Because units, transduction mechanisms, and response linearity differ, values should be interpreted mainly within each modality. Some FBG studies report steady-flow wavelength-shift slopes, whereas others emphasize detection threshold, SNR, or localization instead of an explicit calibration slope.
Response time	Reported only when explicitly defined by the source paper. Values may reflect manual-bending transients, wake-triggered response, or sensor-plus-electronics latency, and are therefore not directly comparable. Our value is measured from the first video-resolved contact event to the first clear FBG deviation from baseline.
SNR	Reported only when explicitly stated. Because definitions vary or are not fully specified, SNR is used as contextual information rather than a strict figure of merit. Our SNR uses $\text{signal} = \mu_{\text{steady}} - \mu_{\text{static}} $ and $\text{noise} = \sigma_{\text{static}}$, expressed in dB.
Approximate values	A leading $\sim$ denotes a value estimated from a plot, local slope, or closely matching secondary summary when not tabulated in the source paper. These values provide context only and are not treated as directly reported author values.

Caption for Movie S1. Towing whiskers of different geometries. Representative towing trials for the three whisker geometries compared in Fig. 4C: wavy tapered, wavy non-tapered, and cylindrical. All three whiskers are towed under the same mounting condition at 0.6 m/s. The cylindrical baseline exhibits the largest VIV, whereas the two wavy whiskers show markedly reduced VIV.

Caption for Movie S2. Towing a whisker at different angles of attack. Representative towing trials for the wavy tapered whisker at AoAs from 0° to 90° in 45° increments, corresponding to Fig. 4E. The whisker is towed at 0.6 m/s while its orientation relative to the incoming flow is changed. Larger oscillatory responses appear as the whisker rotates away from the small-AoA configuration.

Caption for Movie S3. Loading tests across whiskers with different bridge geometries. A linear stage presses laterally against the whisker tip while the applied force and FBG 1 wavelength are recorded synchronously. The video shows representative trials from the five bridge geometries compared during parameter selection: the baseline design (2 mm bridge height, 45 mm bridge length), two shorter-bridge variants (35 and 25 mm at 2 mm bridge height), and two taller-base variants (4 and 6 mm bridge height at 45 mm bridge length).

Caption for Movie S4. Force sensing response and recovery after large loads underwater. A transient load was applied at the marked force-application point to quantify the force sensing response time. The video timing was calibrated using a real-world clock, and the FBG 1 signal onset was found to occur 29.74 ms after contact. The movie corresponds to Fig. S3. Then, the whisker is manually deflected underwater to three force application points and then abruptly released so that it recovers toward its undeformed state. The video plays the three trials in sequence, each overlaid with the corresponding FBG wavelength trace on a common axis so that recovery amplitude and decay time can be compared across loading levels.

Caption for Movie S5. Cyclic loading durability test. The preferred whisker design is subjected to 500 repeated indentation cycles on the linear stage. The video shows representative segments of the test together with the synchronized FBG response, illustrating repeatable loading signatures

over long-duration cycling. This movie corresponds to the durability experiment summarized in Fig. S1.

Caption for Movie S6. Whisker response to a pitching foil in quiescent water. A NACA0012 foil pitches at a fixed location in quiescent water while the whisker is positioned at different points on an x - y grid, one test point per trial. The recorded FBG responses from all grid points are assembled into a spatial map of the disturbance field. This movie is the animated counterpart to Fig. 5D and highlights the laterally distributed response pattern produced in the fixed-pitching quiescent-water case. Each arrow represents the sensor signal from a single test with the foil at the corresponding angle. Hence, the vectors are synchronized in terms of phase but not in simultaneous measurements.

Caption for Movie S7. Translating foil trail analogue. The foil and whisker translate together while the foil continues to pitch, producing the finite- St wake regime used in the branch-selection experiments. The movie assembles the measured whisker responses over the sampled test locations into a map of the disturbance field. This movie is the animated counterpart to Fig. 5E.

Caption for Movie S8. Dual-casting fabrication of the whisker sensor. The video shows the fabrication workflow for the wavy tapered whisker, including positive-mold printing, silicone negative-mold casting, removal of the positive core, nitinol-wire insertion, urethane injection, demolding, and assembly into the FBG base structure.

Caption for Movie S9. Whisker-guided branch selection during delayed trail following on a mobile underwater robot. Representative pool trials in which the robot enters the decision region after the wake source has moved on and then commits to either the straight or turning branch using whisker signals alone. In this setting, the robot's starting point was approximately 0.4 m behind the foil's initial position, and about 5 s elapsed between the onset of foil motion and the robot reaching that position. The movie includes a sliding window of the whisker-sensor signal, illustrating how the FBG response changes as the robot enters the vortex region and how this signal is used to guide branch-selection decisions.

Caption for Movie S10. Field deployment at Hopkins Marine Station. Representative field footage of the whisker-integrated robot at Hopkins Marine Station. The video documents the deployment environment, vehicle operation, and example whisker signals collected during outdoor trials. In the first scenario, the whisker sensor suppresses vortex-induced vibration (VIV), producing stable baseline signals in the ocean environment. Second, the robot detects a wake generated by the diver’s fin, demonstrating whisker-based sensing of field-generated vortical disturbances. Overlaid whisker traces in the two video segments illustrate the contrast between VIV suppression and wake detection under field conditions.